

Multiple Choice (10 marks)

1. Statement 1| RELUs are not monotonic, but sigmoids are monotonic. Statement 2| Neural networks trained with gradient descent with high probability converge to the global optimum.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. What is the dimensionality of the null space of the following matrix? $A = \begin{bmatrix} 3 & 2 & -9 \\ -6 & -4 & 18 \\ 12 & 8 & -36 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
3. Existential risks posed by AI are most commonly associated with which of the following professors?
 - (a) Nando de Freitas
 - (b) Yann LeCun
 - (c) Stuart Russell
 - (d) Jitendra Malik
4. The numerical output of a sigmoid node in a neural network:
 - (a) Is unbounded, encompassing all real numbers.
 - (b) Is unbounded, encompassing all integers.
 - (c) Is bounded between 0 and 1.
 - (d) Is bounded between -1 and 1.
5. A and B are two events. If $P(A, B)$ decreases while $P(A)$ increases, which of the following is true?
 - (a) $P(A|B)$ decreases
 - (b) $P(B|A)$ decreases
 - (c) $P(B)$ decreases
 - (d) All of above

True / False (10 marks)

Answer True or False

6. True or False: Out-of-distribution detection is synonymous with train-test mismatch robustness in machine learning.
7. True or False: The computational complexity of gradient descent is polynomial in the dimensionality D of the data.
8. True or False: The rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ is 2.
9. True or False: Discriminative approaches attempt to model the joint distribution $p(y, x)$, which captures the overall data distribution.
10. True or False: Lasso regression is more suitable for feature selection due to its ability to shrink some coefficients to zero.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the given matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ and determine the dimensionality of its null space. Discuss the implications of your findings in the context of linear transformations.
12. Discuss the conditions under which a linear hard-margin SVM is applicable for training data, and analyze the implications of linear separability in machine learning models.

End of Paper